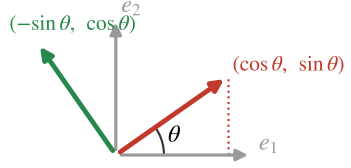
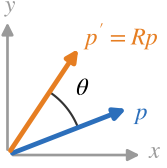
(a) where the basis vectors go

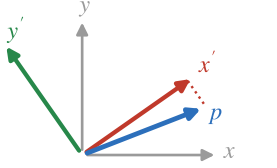
$\Rightarrow$ columns of $R_z(\theta)$



(b) active: the vector moves, the frame stays



(c) passive: the vector stays, the frame moves



coords of p in new frame: $R^{\top} p$